Intangible Investment Restraint and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Intangible Investment Restraint (IIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on IIR achieves an annualized gross (net) Sharpe ratio of 0.43 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (19) bps/month with a t-statistic of 2.85 (2.49), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals) is 19 bps/month with a t-statistic of 2.42.

1 Introduction

Financial markets efficiently incorporate information into asset prices when investors rapidly process and act on new signals. Yet substantial evidence documents systematic deviations from efficiency, particularly in how quickly market participants recognize and respond to complex accounting information embedded in financial statements. We examine whether firms' intangible investment restraint—measured as the negative change in intangible assets scaled by total assets—predicts future stock returns through the lens of gradual information diffusion. This signal captures firms' strategic decisions to limit or reduce intangible capital expenditures, potentially signaling either financial discipline or underinvestment in growth opportunities that market participants fail to immediately recognize.

The slow diffusion of information provides a compelling framework for understanding why intangible investment restraint predicts returns. Hong and Stein (1999) demonstrate that firm-specific information diffuses gradually across the investment community, generating momentum in stock prices as investors sequentially update their beliefs. This gradual incorporation mechanism becomes particularly pronounced for complex accounting signals that require sophisticated analysis to interpret. Hirshleifer and Teoh (2003) show that investors exhibit limited attention to information in financial statements, leading to systematic underreaction to value-relevant accounting measures. The complexity of intangible assets—which include research and development, patents, brand value, and human capital—makes changes in these investments especially susceptible to delayed market recognition.

Intangible investment restraint represents a particularly opaque signal that investors process slowly for several reasons. First, accounting standards provide limited disclosure requirements for intangible investments, creating information frictions that impede rapid price discovery. Lev (2001) document that the accounting treatment of intangibles creates systematic information asymmetries between managers and outside investors. Second, evaluating whether reduced intangible investment reflects prudent capital allocation or myopic underinvestment requires detailed industry knowledge and forward-looking analysis that many investors lack the resources to conduct immediately. Third, the benefits or costs of intangible investment decisions typically materialize over extended horizons, making their immediate valuation implications ambiguous.

The gradual diffusion hypothesis predicts that returns to intangible investment restraint strategies should concentrate among stocks with greater information frictions. Zhang (2006) shows that information uncertainty amplifies the magnitude of return predictability from accounting-based signals. Small firms, those with low analyst coverage, and stocks with limited institutional ownership face higher barriers to information transmission, creating environments where prices adjust more slowly to fundamental news. DellaVigna and Pollet (2009) provide evidence that investor inattention varies predictably across firms and time periods, with neglected stocks exhibiting stronger return continuation patterns. These cross-sectional predictions provide testable implications for where intangible investment restraint should generate the strongest return predictability.

Our empirical analysis reveals that intangible investment restraint strongly predicts cross-sectional stock returns. The value-weighted long-short strategy that buys stocks with high IIR and sells those with low IIR generates an average monthly return of 23 basis points with a t-statistic of 3.05, translating to an annualized Sharpe ratio of 0.43. The strategy's alpha remains economically and statistically significant across all standard factor models, with the six-factor alpha equaling 22 basis points per month (t-statistic = 2.85). These results demonstrate that IIR captures a distinct source of return predictability not explained by established risk factors including market, size, value, profitability, investment, and momentum.

The economic magnitude of the IIR effect remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.35 places the strategy in the 93rd percentile among 212 documented anomalies, while the net six-factor alpha of 19 basis points per month (t-statistic = 2.49) confirms that implementation frictions do not eliminate the strategy's profitability. Notably, the strategy maintains significant performance among large-cap stocks, generating an average return of 30 basis points per month (t-statistic = 2.74) in the highest size quintile. This large-stock performance suggests that IIR captures systematic mispricing rather than merely compensating for illiquidity or microstructure effects that plague many anomalies.

Controlling for closely related anomalies barely attenuates the IIR signal's predictive power. When we simultaneously control for the six most correlated predictors—Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, and Accruals—plus the six Fama-French factors, the IIR strategy retains an alpha of 19 basis points per month with a t-statistic of 2.42. The strategy's gross Sharpe ratio of 0.43 exceeds 83% of anomalies in the factor zoo, while a dollar invested in the IIR strategy would have grown to $2.62, ranking in the top 8% of all documented anomalies. These results establish that intangible investment restraint provides incremental information about expected returns beyond existing predictors.

This paper contributes to three distinct literatures in asset pricing and accounting. First, we extend the investment-based asset pricing literature by documenting that intangible investment restraint predicts returns with magnitude comparable to tangible asset growth. While Cooper et al. (2008) show that asset growth negatively predicts returns and Titman et al. (2004) document similar patterns for capital investments, our results reveal that constraints on intangible investments generate distinct pricing implications not subsumed by traditional investment measures. Unlike physical capital expenditures that appear immediately on balance sheets, intangible investment restraint reflects strategic decisions about innovation, human capital, and organizational capabilities that investors recognize only gradually.

Second, our findings contribute to the literature on limited attention and gradual information diffusion in financial markets. Building on Hirshleifer et al. (2009) who demonstrate that investors systematically neglect information in financial statements, we identify intangible investment restraint as a particularly opaque signal subject to delayed market recognition. The concentration of returns among large stocks distinguishes IIR from other accounting anomalies that primarily manifest in small, illiquid securities. This pattern supports models of rational inattention where sophisticated investors gradually acquire and process complex information rather than market segmentation explanations based on institutional frictions.

Third, we advance the accounting literature on intangible assets and their valuation implications. Lev and Sougiannis (1996) argue that accounting treatment of intangibles creates systematic mispricing, while our results provide direct evidence that changes in intangible investments predict returns through gradual information incorporation. The persistence of the IIR effect after controlling for standard risk factors and related anomalies suggests that current accounting standards inadequately convey the economic implications of intangible investment decisions. These findings have implications for financial statement analysis, suggesting that investors who develop expertise in evaluating intangible investment patterns can generate substantial abnormal returns even in the presence of sophisticated competition.

2 Data

We obtain accounting data from the COMPUSTAT North America database, which provides comprehensive financial statement information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-listed firms and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature.signal construction relies on two key accounting variables. Increase in investments (IVCH) represents the annual change in a firm's investment account, capturing year-over-year changes in the firm's holdings of equity investments, debt securities, and other investment assets reported on the balance sheet. Intangible assets (INTAN) measures the book value of a firm's intangible assets net of goodwill, including items such as patents, trademarks, licenses, and other intellectual property. Both variables are reported in millions of dollars and reflect fiscal year-end values from firms' annual financial statements.construct the Intangible Investment Restraint signal by computing the year-over-year change in increase in investments, scaled by lagged intangible assets. Specifically, we calculate (IVCH(t) - IVCH(t-1)) / INTAN(t-1), where t denotes the fiscal year. This measure captures the acceleration or deceleration in a firm's investment activity relative to its intangible asset base. A positive value indicates accelerating investment growth scaled by the firm's existing intangible resources, while a negative value suggests restraint or deceleration in investment expansion. We require non-missing values for IVCH in consecutive years and positive values for lagged INTAN to ensure meaningful scaling. Following standard practice, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IIR signal. Panel A plots the time-series of the mean, median, and interquartile range for IIR. On average, the cross-sectional mean (median) IIR is -0.39 (0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input IIR data. The signal's interquartile range spans -0.01 to 0.01. Panel B of Figure 1 plots the time-series of the coverage of the IIR signal for the CRSP universe. On average, the IIR signal is available for 3.74% of CRSP names, which on average make up 4.35% of total market capitalization.

4 Does IIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IIR portfolio and sells the low IIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IIR strategy earns an average return of 0.23% per month with a t-statistic of 3.05. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.22% to 0.27% per month and have t-statistics exceeding 2.85 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 4.51 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 372 stocks and an average market capitalization of at least $1,079 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.32. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -8-29bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.16. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the IIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IIR, as well as average returns and alphas for long/short trading IIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80$^{\text{th}}$ NYSE percentile), the IIR strategy achieves an average return of 30 bps/month with a t-statistic of 2.74. Among these large cap stocks, the alphas for the IIR strategy relative to the five most common factor models range from 30 to 37 bps/month with t-statistics between 2.65 and 3.31.

5 How does IIR perform relative to the zoo?

Figure 2 puts the performance of IIR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the IIR strategy falls in the distribution. The IIR strategy's gross (net) Sharpe ratio of 0.43 (0.35) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IIR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the IIR strategy would have yielded \$2.62 which ranks the IIR strategy in the top 8% across the 212

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

anomalies. Accounting for trading costs, a $1 invested in the IIR strategy would have yielded $1.85 which ranks the IIR strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IIR relative to those. Panel A shows that the IIR strategy gross alphas fall between the 49 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases IIR ranks between the 67 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IIR with 208 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IIR or at least to weaken the power IIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IIR conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IIR} IIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IIR. Stocks are finally grouped into five IIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IIR trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IIR signal in these Fama-MacBeth regressions exceed -0.65, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on IIR is -0.45.

Similarly, Table 5 reports results from spanning tests that regress returns to the

an average month.

IIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IIR strategy earns alphas that range from 21-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.71, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IIR trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.42.

7 Does IIR add relative to the whole zoo?

Finally, we can ask how much adding IIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the IIR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes IIR grows to $1083.82.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IIR is available.

8 Conclusion

This study provides robust evidence that Intangible Investment Restraint (IIR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IIR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.35, which remains economically significant after accounting for transaction costs. Most notably, the strategy delivers a monthly alpha of 19 basis points (t-statistic = 2.42) even after controlling for the Fama-French five factors, momentum, and six closely related strategies from the factor zoo, suggesting that IIR captures unique information about future returns not subsumed by existing factors.

The practical implications of these findings are considerable for both academic researchers and investment practitioners. The persistence of IIR's predictive power after controlling for related financing and investment strategies—including net external financing, asset growth, and accruals—indicates that constraints on intangible investment represent a distinct channel through which firm characteristics influence expected returns. This suggests that investors can potentially enhance portfolio performance by incorporating IIR signals into their investment processes, particularly given the strategy's robustness to transaction costs as evidenced by the modest decline from gross to net Sharpe ratios.

Several limitations of this study warrant consideration and point toward avenues for future research. First, while our analysis demonstrates statistical significance, the economic magnitude of returns, though meaningful, may be subject to implementation challenges in practice, including market impact costs not fully captured in our transaction cost estimates. Second, the study period and market conditions examined may not fully represent the signal's performance across different market regimes or in international markets. Future research should investigate the performance of IIR in different geographical markets and examine whether the predictive power varies across different economic cycles or market conditions. Additionally, researchers should explore the underlying economic mechanisms driving the IIR anomaly, potentially examining whether it reflects systematic risk compensation or market inefficiencies related to investor inattention to intangible investment patterns. Finally, investigating the interaction between IIR and firm-specific characteristics such as size, industry classification, and intangible asset intensity could provide deeper insights into when and where this signal is most effective.

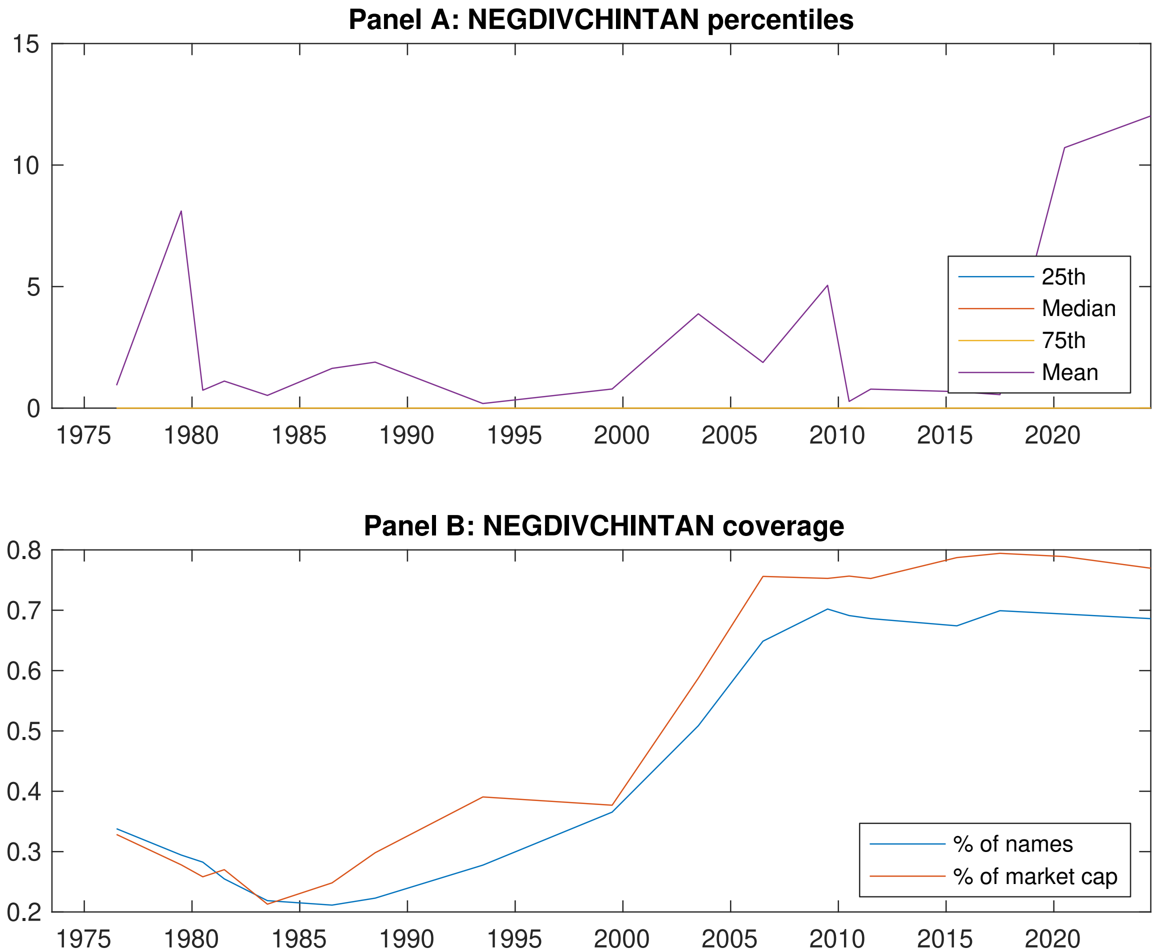


Figure 1: Times series of IIR percentiles and coverage.
This figure plots descriptive statistics for IIR. Panel A shows cross-sectional percentiles of IIR over the sample. Panel B plots the monthly coverage of IIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on IIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.68	0.70	0.73	0.77	0.23
	[2.72]	[3.42]	[3.46]	[3.76]	[3.95]	[3.05]
α_{CAPM}	-0.13	0.02	0.04	0.09	0.11	0.24
	[-2.21]	[0.31]	[0.53]	[1.33]	[1.96]	[3.18]
α_{FF3}	-0.17	0.00	0.05	0.06	0.09	0.27
	[-3.04]	[0.01]	[0.65]	[0.97]	[1.64]	[3.47]
α_{FF4}	-0.13	0.01	0.05	0.08	0.12	0.26
	[-2.32]	[0.17]	[0.61]	[1.16]	[2.12]	[3.28]
α_{FF5}	-0.18	-0.14	-0.11	-0.09	0.05	0.22
	[-3.09]	[-2.15]	[-1.48]	[-1.45]	[0.79]	[2.90]
α_{FF6}	-0.15	-0.12	-0.10	-0.07	0.08	0.22
	[-2.56]	[-1.82]	[-1.34]	[-1.08]	[1.29]	[2.85]
Panel B: Fama and French (2018) 6-factor model loadings for IIR-sorted portfolios						
β_{MKT}	1.03	1.04	1.02	1.01	1.03	0.00
	[77.24]	[68.54]	[58.99]	[72.00]	[75.75]	[0.02]
β_{SMB}	-0.01	0.09	0.13	0.12	-0.05	-0.04
	[-0.49]	[3.91]	[4.84]	[5.64]	[-2.37]	[-1.42]
β_{HML}	0.14	-0.10	-0.13	-0.06	-0.02	-0.15
	[5.30]	[-3.31]	[-3.99]	[-2.23]	[-0.65]	[-4.39]
β_{RMW}	0.08	0.24	0.36	0.32	0.04	-0.04
	[3.11]	[8.03]	[10.73]	[11.64]	[1.50]	[-1.16]
β_{CMA}	-0.07	0.26	0.14	0.17	0.17	0.24
	[-1.85]	[5.88]	[2.71]	[4.24]	[4.20]	[4.51]
β_{UMD}	-0.05	-0.03	-0.01	-0.03	-0.05	0.00
	[-3.63]	[-2.18]	[-0.84]	[-2.49]	[-3.43]	[0.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	372	404	425	412	374	
me ($\$10^6$)	2666	1134	1079	1111	2495	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23	0.24	0.27	0.26	0.22	0.22
			[3.05]	[3.18]	[3.47]	[3.28]	[2.90]	[2.85]
Quintile	NYSE	EW	0.13	0.13	0.12	0.13	0.08	0.09
			[2.32]	[2.33]	[2.12]	[2.14]	[1.46]	[1.60]
Quintile	Name	VW	0.20	0.21	0.23	0.22	0.19	0.19
			[2.70]	[2.84]	[3.09]	[2.87]	[2.52]	[2.45]
Quintile	Cap	VW	0.33	0.35	0.39	0.39	0.32	0.33
			[3.36]	[3.51]	[3.97]	[3.86]	[3.20]	[3.21]
Decile	NYSE	VW	0.25	0.26	0.30	0.33	0.30	0.32
			[2.20]	[2.28]	[2.58]	[2.79]	[2.48]	[2.69]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.19	0.20	0.23	0.22	0.20	0.19
			[2.54]	[2.66]	[2.97]	[2.91]	[2.57]	[2.49]
Quintile	NYSE	EW	-0.08					
			[-1.16]					
Quintile	Name	VW	0.16	0.17	0.19	0.19	0.16	0.15
			[2.17]	[2.33]	[2.60]	[2.53]	[2.19]	[2.09]
Quintile	Cap	VW	0.29	0.30	0.34	0.34	0.29	0.28
			[2.96]	[2.98]	[3.40]	[3.38]	[2.83]	[2.78]
Decile	NYSE	VW	0.21	0.20	0.23	0.25	0.23	0.24
			[1.84]	[1.72]	[2.04]	[2.21]	[1.99]	[2.07]

Table 3: Conditional sort on size and IIR

This table presents results for conditional double sorts on size and IIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IIR and short stocks with low IIR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

	IIR Quintiles					IIR Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.78	0.93	0.81	0.88	0.90	0.11	0.09	0.10	0.10	0.09	0.09
	[2.95]	[3.14]	[2.89]	[3.18]	[3.30]	[1.14]	[0.94]	[0.95]	[0.94]	[0.87]	[0.90]
(2)	0.88	0.86	0.88	0.84	0.89	0.01	0.02	0.03	0.01	-0.06	-0.07
	[3.32]	[3.29]	[3.43]	[3.32]	[3.48]	[0.09]	[0.21]	[0.27]	[0.05]	[-0.54]	[-0.62]
(3)	0.72	0.92	0.81	0.90	0.80	0.08	0.10	0.09	0.03	0.02	-0.02
	[2.96]	[3.93]	[3.42]	[3.72]	[3.37]	[0.82]	[0.98]	[0.93]	[0.31]	[0.17]	[-0.22]
(4)	0.84	0.79	0.81	0.81	0.77	-0.08	-0.06	-0.08	-0.04	-0.17	-0.13
	[3.72]	[3.50]	[3.52]	[3.65]	[3.44]	[-0.78]	[-0.61]	[-0.84]	[-0.38]	[-1.68]	[-1.25]
(5)	0.44	0.63	0.60	0.70	0.74	0.30	0.32	0.35	0.37	0.30	0.32
	[2.13]	[3.25]	[3.04]	[3.79]	[3.68]	[2.74]	[2.94]	[3.21]	[3.31]	[2.65]	[2.81]

Panel B: Portfolio average number of firms and market capitalization

	IIR Quintiles Average n					IIR Quintiles Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	207	205	205	206	206	26	23	22	22	25
(2)	67	67	67	67	66	47	47	46	45	47
(3)	48	48	48	48	48	84	86	84	83	84
(4)	40	40	40	40	40	182	184	178	182	183
(5)	37	37	37	37	37	1619	1259	1149	1153	1624

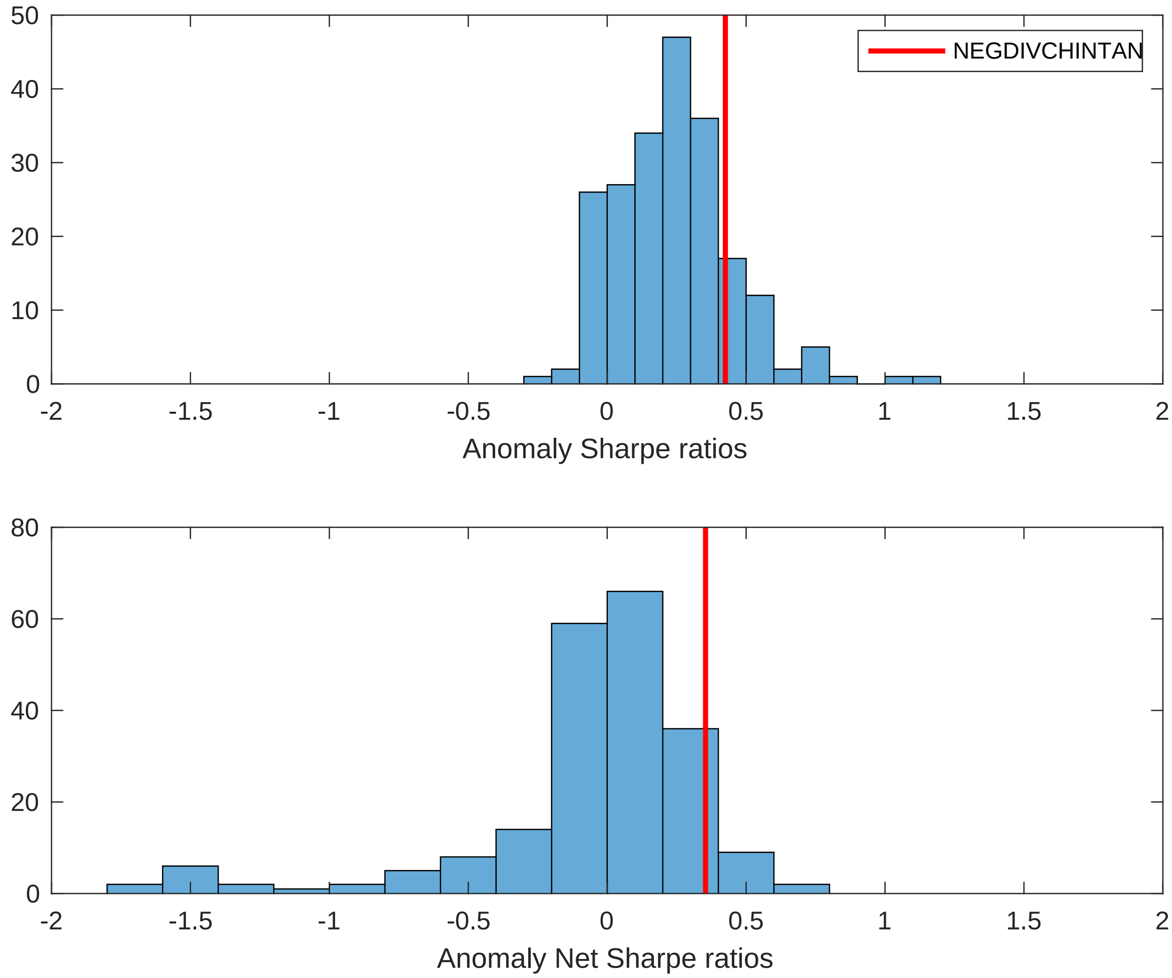


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

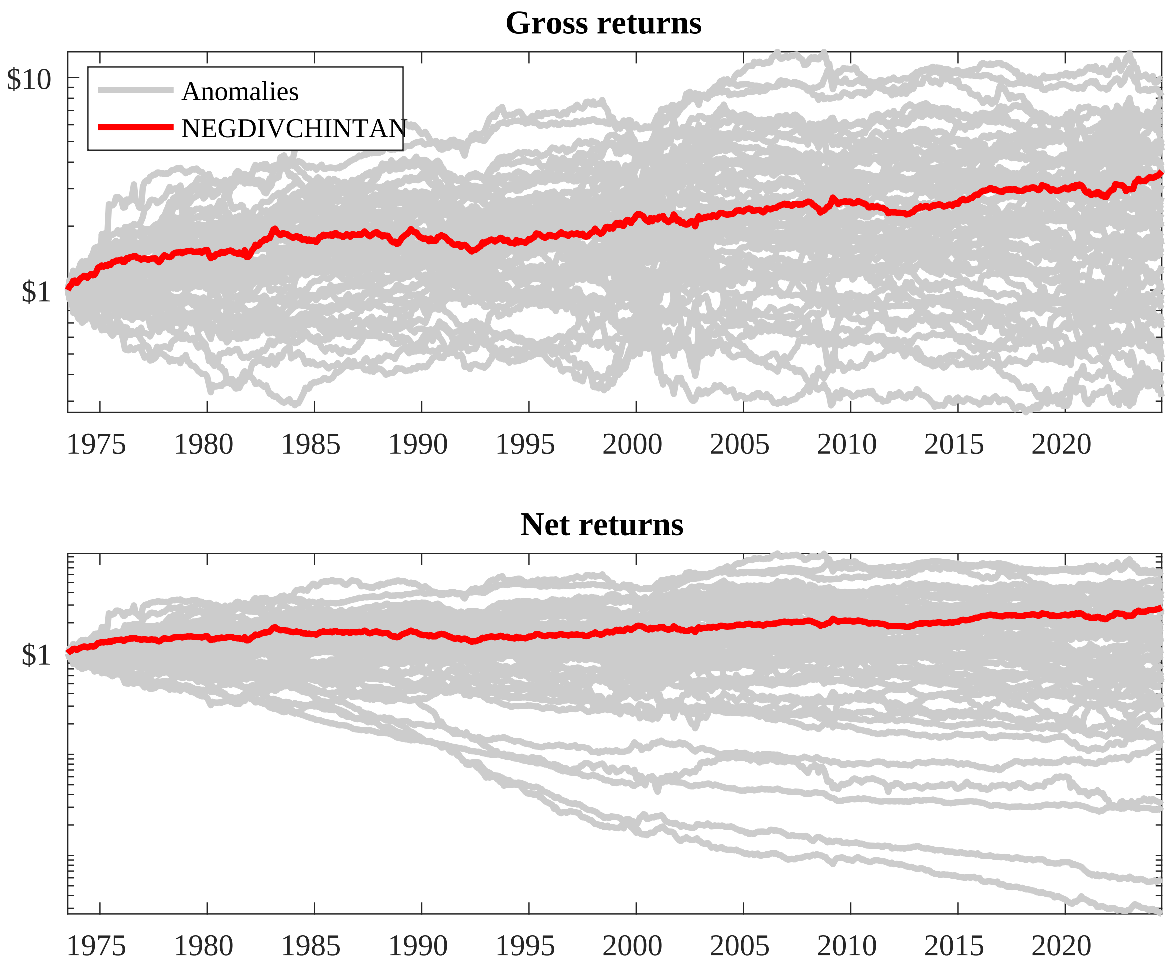
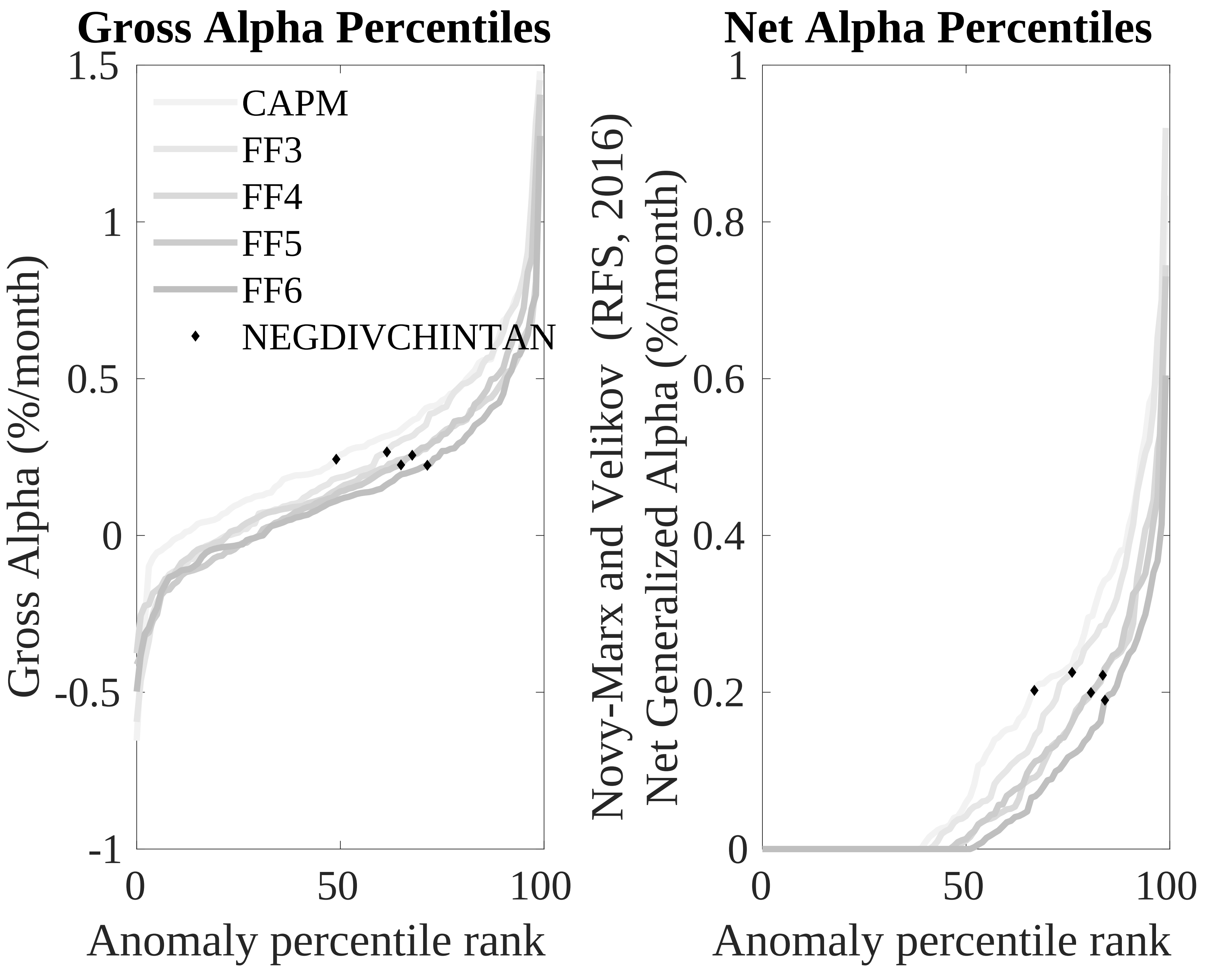


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDIVCHINTAN
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank

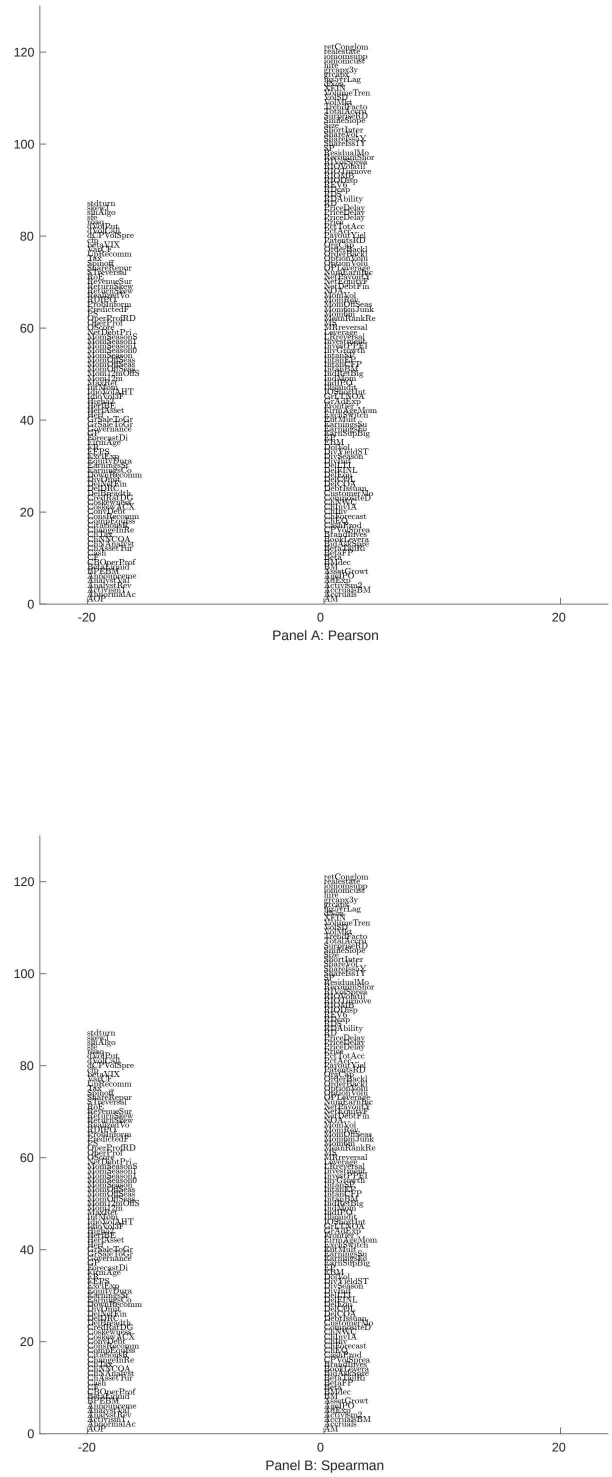


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 208 filtered anomaly signals with IIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

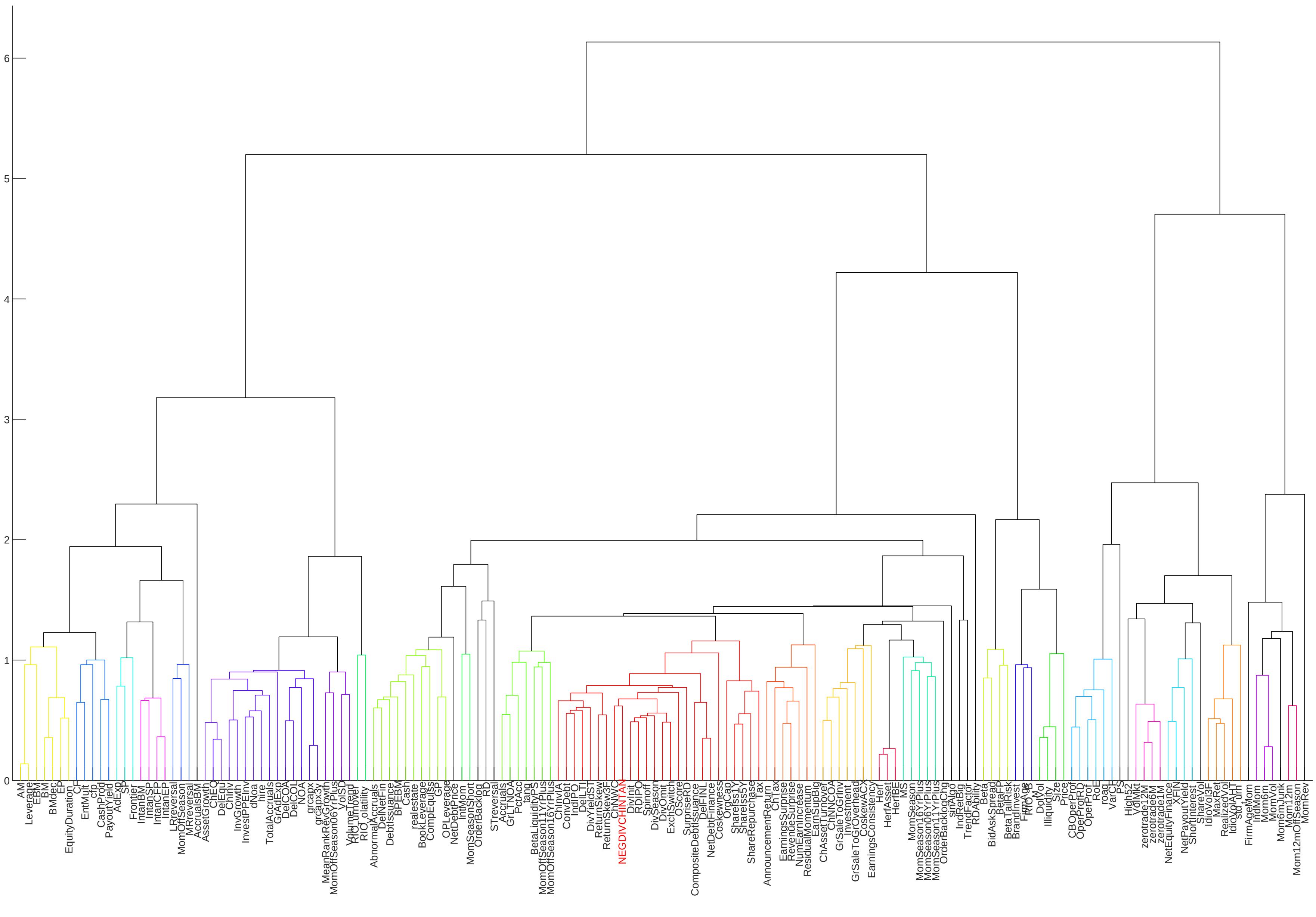


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

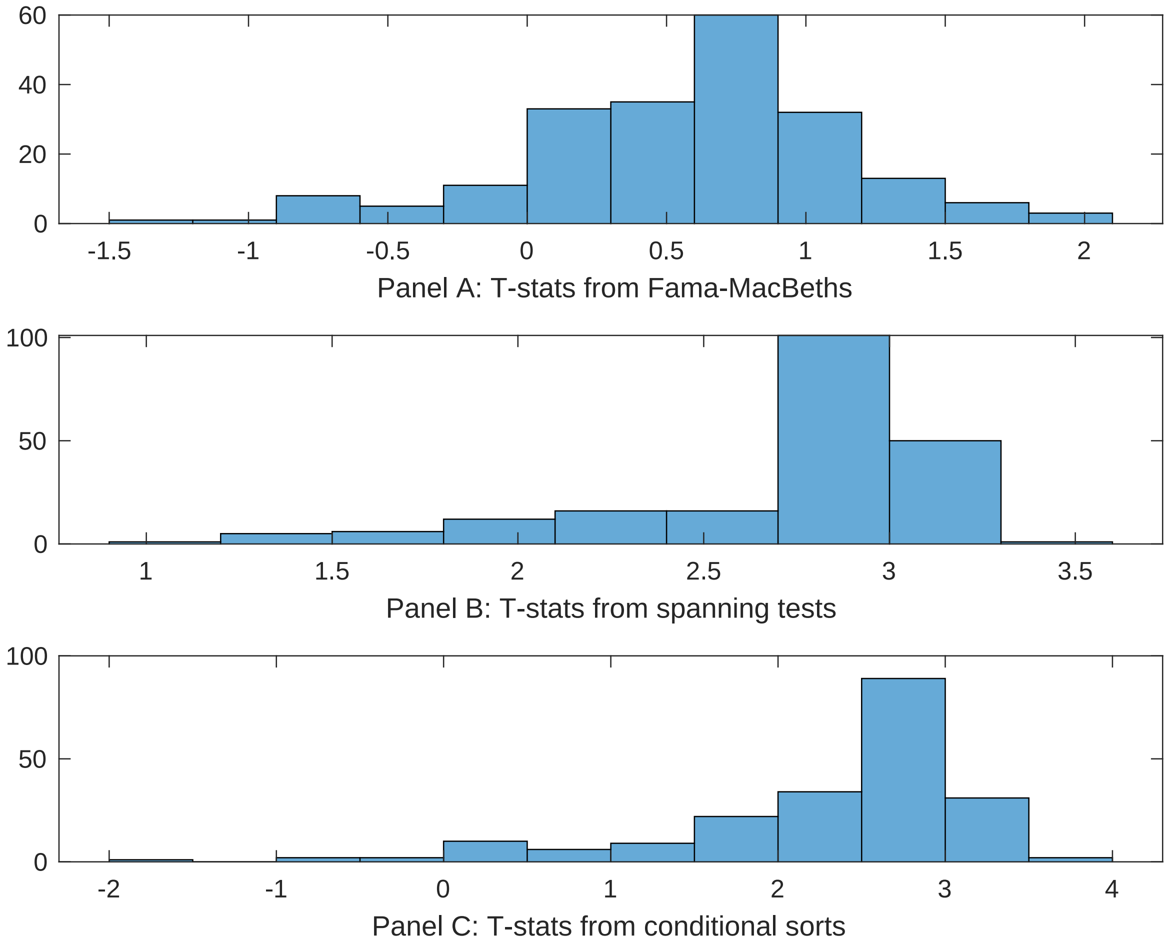


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of IIR conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IIR} IIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IIR. Stocks are finally grouped into five IIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IIR trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IIR} IIR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.13	0.13	0.14	0.12	0.14
	[5.52]	[5.28]	[5.28]	[5.18]	[5.70]	[4.96]	[5.51]
IIR	0.17	0.26	0.64	0.16	-0.38	0.32	-0.28
	[0.26]	[0.41]	[0.11]	[0.25]	[-0.65]	[0.51]	[-0.45]
Anomaly 1	0.21						0.13
	[6.91]						[1.98]
Anomaly 2		0.20					-0.20
		[8.16]					[-0.29]
Anomaly 3			0.15				-0.28
			[3.86]				[-0.54]
Anomaly 4				0.45			0.13
				[1.86]			[0.60]
Anomaly 5					0.99		0.69
					[8.00]		[5.16]
Anomaly 6						0.17	0.75
						[4.74]	[2.37]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IIR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net external financing, Net debt financing, Change in equity to assets, Total accruals, Asset growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21	0.21	0.22	0.22	0.22	0.21	0.19
	[2.71]	[2.78]	[2.87]	[2.81]	[2.87]	[2.72]	[2.42]
Anomaly 1	11.98						9.10
	[3.09]						[2.19]
Anomaly 2		11.43					9.52
		[2.70]					[2.14]
Anomaly 3			9.23				10.02
			[2.41]				[2.04]
Anomaly 4				4.54			1.86
				[1.29]			[0.47]
Anomaly 5					6.63		-2.70
					[1.47]		[-0.52]
Anomaly 6						3.98	4.67
						[1.29]	[1.50]
mkt	1.59	-0.03	-0.13	-0.20	-0.06	0.24	1.46
	[0.86]	[-0.02]	[-0.07]	[-0.11]	[-0.03]	[0.14]	[0.78]
smb	-1.31	-6.17	-5.12	-4.97	-5.64	-4.40	-1.89
	[-0.44]	[-2.27]	[-1.90]	[-1.83]	[-2.07]	[-1.59]	[-0.59]
hml	-15.71	-16.57	-18.27	-16.83	-17.34	-16.02	-15.64
	[-4.55]	[-4.82]	[-5.25]	[-4.87]	[-5.02]	[-4.53]	[-4.37]
rmw	-10.97	-5.07	-2.98	-3.01	-3.79	-2.50	-7.54
	[-2.62]	[-1.44]	[-0.85]	[-0.85]	[-1.08]	[-0.69]	[-1.73]
cma	17.55	22.39	16.17	22.72	17.33	24.07	7.71
	[3.03]	[4.20]	[2.50]	[4.04]	[2.29]	[4.51]	[0.98]
umd	-0.24	-0.90	-0.09	-0.02	-0.02	-0.46	-1.13
	[-0.14]	[-0.50]	[-0.05]	[-0.01]	[-0.01]	[-0.26]	[-0.63]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	6	6	6	5	5	5	8

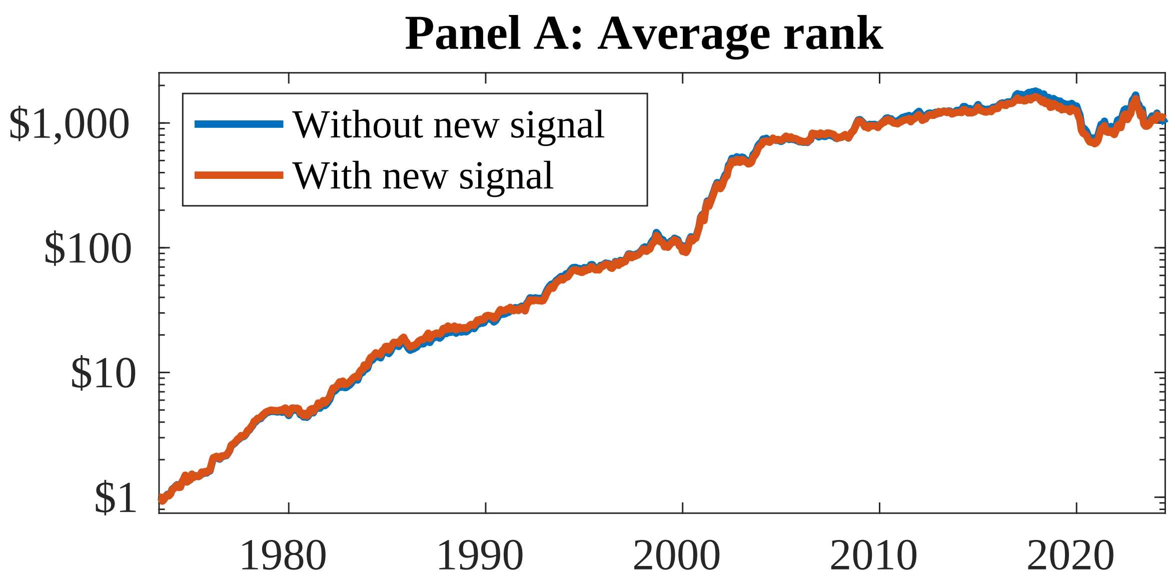


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as IIR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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